

JOON KIM

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EDUCATION

The Cooper Union, Albert Nerken School of Engineering

New York, NY

Bachelor of Science in Mechanical Engineering [Minor in Philosophy]

Class of 2028

- **GPA:** 3.85/4.0 | Cooper Union Merit Scholar
- **Relevant Coursework:** Feedback Control Systems, Thermodynamics, System Dynamics, Circuits, Fluid Mechanics, Stress, Design & Prototyping, Mechanics of Materials, Modern Physics, Materials Science, Principles of Design, Differential Equations

PROFESSIONAL EXPERIENCE

MORA

Seoul, South Korea

Founder & Lead Engineer

May 2026 – Present

- Founded MORA Robotics, an ag-tech startup, and led development of an autonomous field-transport robot for Korea's narrow dry-field rows, floor-testing a full follow-collect-dump-return loop on real hardware with vision-based farmer tracking.
- Built and validated a fully functional first prototype with a tracked, tread-wheel chassis featuring load-leveling suspension, onboard computer vision, and distance-sensing, proving the platform's core mobility and sensing systems on real hardware.
- Pitched early-stage VCs and investors including Krew Capital, connected with robotics engineers, and informally advised first-time founders, relationships built off a LinkedIn presence for MORA that drew 100K+ views. | [\[Company Website\]](#)

Cooper Union Dynamics and Control Lab

New York, NY

Lead Research Scholar

December 2022 – August 2024, September 2026 – Present

- Secured \$7,000+ in research funding and created an interactive educational platform adopted by the university lab for undergraduate demonstrations, workshops, and annual showcases.
- Developed intuitive hand-tracking control systems using OpenCV and MediaPipe to translate gestures into real-time drone swarm simulations of complex dynamic systems, including mass-spring-damper and pendulum models.
- Streamlined the data communication pipeline across adjacent research teams within the VIP drone program, resolving a critical 2-second latency issue to enable precise, low-latency control. | [\[Portfolio Page\]](#)

Hyundai Rotem and Republic of Korea Army

Cheorwon County, South Korea

Lead Armored Vehicle Specialist

October 2024 – April 2026

- Led a team of 5 in the maintenance and repair of over 35 armored vehicles, executing a wide scope of services from routine filter and tire changes to complex engine modifications and armor restorations.
- Oversaw fleet logistics by managing both the maintenance history database and a 50+ unique part physical inventory, ensuring data accuracy while optimizing stock levels to guarantee parts availability.
- Directed four inter-unit training sessions, guiding external teams and officers on specialized maintenance techniques and repair procedures for armored personnel vehicles.

Real Cube Co., Ltd.

Seoul, South Korea

Product Design and Prototyping Intern

June 2023 – July 2023

- Collaborated with cross-functional engineers and researchers to design and prototype innovative air purifier filter shells, contributing to the final design of the "Airwell 99" air purifier.
- Designed 10+ detailed 3D models of filter shells using Fusion 360 and produced high-fidelity prototypes utilizing 3D printers.
- Performed quality control and troubleshooting on prototypes, identifying and resolving 15+ design flaws to ensure optimal performance and design integrity.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Cooper Union VIP IGVC Autonomous Vehicles

New York, NY

Controls Team

November 2022 – December 2023

- Developed a ROS navigation stack using 3D LiDAR data for real-time SLAM, object detection, and path planning.
- Implemented and tuned PID controllers for steering and throttle control, reducing steering error by 30%.
- Introduced a LiDAR-based perception system that was instrumental in winning the IGVC Innovation Award.

Cooper Union Cooper Consulting Group (CCG)

New York, NY

Consulting Development Lead

September 2023 – May 2024

- Ran financial modeling and market analysis workshops and logistics for 20+ members, raising semester attendance by 25%.
- Presented business solutions at the Global Case Competition at Harvard (GCCH) with an 8-person multidisciplinary team.

SKILLS & INTERESTS

- **Skills:** Programming (Python, C, MATLAB, ROS, OpenCV, MediaPipe), CAD Software (Fusion 360, Onshape, AutoCAD), Fabrication Skills (3D Printing, CNC Milling/Programming, Drill Press Operation, Woodworking), Microsoft Office Suite, AI Tools (Claude Code, AI-assisted Coding, Building Custom Agents and Workflows)
- **Languages:** English (Native Fluent), Korean (Native Fluent), Spanish (Working Proficiency), Chinese (Working Proficiency)
- **Interests:** Controls Systems, Robotics, Product Design, Golfing, Freestyle Wrestling, Weightlifting, Piano Jazz Composition